Unit 1
Human–Computer Interaction

HCI · accessibility · UI/UX · wireframes · Figma · evaluate

TRADITIONAL VS NATURAL | CONTRAST & INCLUSIVE NAV | LOW-FI → HIGH-FI | SUS & NPS

If a human cannot use it, the code does not matter. HCI is the subject that says that out loud.

0. HCI in one sentence

Human–Computer Interaction (HCI) — also called CHI or MMI (Man–Machine Interaction) — is the design of technology so that it is easy, efficient, and respectful of human bodies and minds. UI is what you see; UX is what you feel across the whole journey.

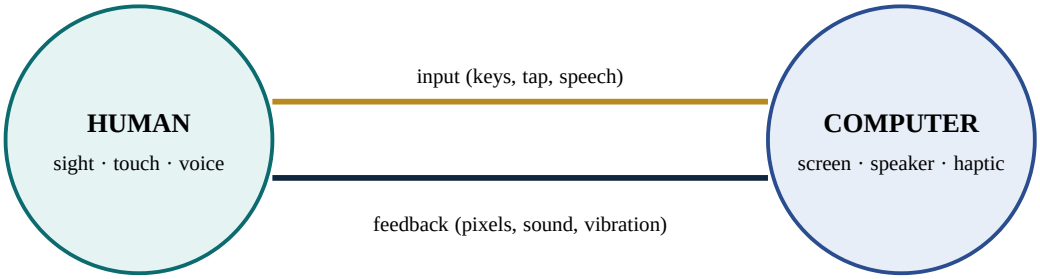


Figure 1. The HCI loop. A missing feedback channel is a defect, not a style choice.

0.1 Sensory channels

SIGHT
Screens, icons, video

TOUCH
Tap, type, swipe ↔ click or vibration

HEARING
Spoken prompts, notification sounds

VOICE
“Alexa, play music”

SPATIAL
Body position in a motion game / VR

Table 1. Two families of interaction

Traditional (Interaction I)	Natural (Interaction II)
Mouse, keyboard, tap/swipe	Voice, gesture, face, spatial tracking
PC: keys + mouse → monitor + speakers	Smart speaker, Kinect, face unlock
Phone: tap/swipe → light, sound, vibrate	Multimodal: several channels at once

1. Components, types, and why HCI pays

Components: user (end-user vs stakeholder), computer system, interaction, task, UI, environment, feedback (system ↔ user).

Interaction styles: direct manipulation, command line, menus, forms, conversation, gesture, voice, context-aware, biometric, haptic, multimodal.

UI families: GUI, CLI, touch, VUI, NUI (VR / Kinect).

Why it matters: faster work, inclusion, room for AR/VR/AI/IoT, commercial survival, ethical duty not to exclude.

2. Accessibility is not a theme — it is a constraint

CONTRAST
Text must stand off the background. Pale grey on white fails in sunlight and for low vision.

READABILITY
Sane type size, line length, and language. Decorative fonts are not “friendly”.

COLOUR-BLIND SAFE
Never rely on colour alone. Pair red/green with a symbol and a label.

INCLUSIVE NAVIGATION
Tab order, visible focus, screen-reader names, captions, vibration as a second channel.



Figure 2. Accessibility tests you can do on paper: contrast, and “would this still work in greyscale?”

3. Need analysis — design for someone, not for “users”

A **need** is the gap between the current state and the desired state. Steps: identify users → collect information → analyse → prioritise → propose solutions.

Mobile banking example: biometric login, transfers, an elderly-friendly large-type mode, and a second language are needs, not decorations. A 19-year-old designer who only tests on their own phone will miss all four.

4. Common HCI failures and how you repair them

Table 2. Problems you will be asked to name

Symptom	Likely cause	Repair
Users cannot find the next button	Hidden navigation, poor hierarchy	Visible primary action, consistent menu
Errors with no explanation	System-centred messages	Plain language + how to fix
Tiny hit targets	Desktop layout on a phone	Thumb-sized targets, spacing
Colour-only status	Ignored accessibility	Icon + text
Ten-step signup	No need analysis	Cut fields; progressive disclosure

Improvement methods: user-centred design, iterative prototypes, heuristic review, captions and keyboard paths, A/B tests, watch a real person attempt a real task in silence while you take notes.

5. UI, UX, wireframe, prototype

UI
Visual and interactive layout: colour, type, controls, spacing.

UX
The whole experience: ease, trust, time-to-task, how you feel when you leave.

A **wireframe** is a layout blueprint. A **prototype** is an early model you can tap to test flow before anyone writes production code.

Table 3. Fidelity

	Low-fidelity	High-fidelity
Looks like	Boxes, lines, labels	Colour, type, images, motion
Purpose	Structure and flow, cheap changes	Look-and-feel, stakeholder demo
Speed	Minutes (paper or Figma frames)	Hours to days

5.1 Figma lab — the course tool

Figma is a browser-based collaborative UI tool. Typical class path: **E-Tutorial of HCI** or a **college attendance app**.

1. New file. Press **F** (Frame). Drag a phone frame. Set width/height in the right pane.
2. Duplicate frames for Welcome, Sign-up, Home, Attendance mark, Confirm.
3. Place boxes for title, fields, and a single obvious primary button.
4. Prototype tab: connect buttons to the next frame. Present and tap through.
5. Watch one classmate use it without help. Note every pause. That pause is a defect.

6. Test and evaluate

SUS
System Usability Scale — a short questionnaire, score out of 100.

NPS
Net Promoter Score — "Would you recommend this?"

TASK SUCCESS
Did the user complete the job, and in how many seconds / errors?

Evaluation kinds: user-centred, formative (during design), summative (after), expert review, survey/interview, accessibility audit.

Testing kinds: usability, task-based, automated audit, keyboard/screen-reader walkthrough, prototype test, A/B, compatibility, performance/load.

Pipeline the textbook likes: design → prototype → user test → feedback → final attendance app.

Self-check

1. UI or UX: “The transfer took four taps and I was never unsure”?
2. Why is a red-only error message an accessibility defect?
3. Name one formative and one summative activity.

Answers. (1) UX (the journey), though UI helped. (2) Colour-blind and greyscale users lose the signal; add a word and an icon. (3) Formative: paper-prototype test mid-week. Summative: SUS after the finished Figma flow.

Recap

1. HCI = humans + computers + the loop of input and feedback.
2. Traditional (keys/mouse/tap) vs natural (voice/gesture/spatial).
3. Accessibility: contrast, type, colour+symbol, keyboard/screen reader.
4. Need analysis before pixels. UI $\neq$ UX. Low-fi before high-fi.
5. Figma: frames, prototype links, watch a real user.
6. Evaluate with SUS, NPS, task success; test early and again at the end.

Sketch a 4-frame attendance wireframe on paper, then rebuild it in Figma.